Data & Graph Construction

Validated Temporal Analysis

Counterfactual Layer

Network-flow datasets

APEX-IDS2026
69.1 M flows · 44 days

CIC-IDS2017
20,000 flows · 10 snapshots

UNSW-NB15
flow-level baseline only

Timestamp sort & snapshot partition

Temporal graph snapshots

G_1 G_2 G_3 ⋮ G_T

Topology Genesis Engine

$\Delta_t = (E_t^+, E_t^-, V_t^+, V_t^-)$

$E_t^+ = E_{t+1} \setminus E_t$

$E_t^- = E_t \setminus E_{t+1}$

overlay of G_t and G_{t+1}

persisting edge · added edge E^+ · removed edge E^-

benign training snapshots · *snapshot sequence* · *Δ targets* · *transition statistics*

Topology grammar evolution

Benign training snapshots

Benign grammar (edge whitelist)

$R = \bigcup E_t,\ t \in \text{benign}$

Whitelist check: $e \in R$?

process-novelty flag

Temporal edge prediction (EvolveGCN-H)

history $G_{1:t}$

GRU

GCN

Edge scorer

next-snapshot edge existence

$W_{t+1} = \text{GRU}(Z_t, W_t)$

Leakage control: negatives drawn only from nodes already present in the current graph

Persistence baseline

$\hat{E}_{t+1} = E_t$

compared on APEX-IDS2026 and CIC-IDS2017

Topology anomaly detection

$\tau = \mu_{train} + \sigma_{train}$

train · test

edge-addition statistic of Δ per snapshot

attack-associated flags

novelty and anomaly signals

Implemented as research modules — no quantitative results reported

Genesis Violation
(causal interpretation)
[UNVALIDATED]

RGAR
Reverse Genesis Attack Reconstruction
[UNVALIDATED]

CGB
Counterfactual Genesis Barrier
[UNVALIDATED]

Adaptive Defense
(interface only)
[UNVALIDATED]

Validation requires ordered causal attack ground truth and an authorized replay / intervention testbed, which the passive datasets used here (CIC-IDS2017, UNSW-NB15, APEX-IDS2026) do not provide.

empirically evaluated component · implemented, unvalidated